MATH 152 **QUIZ 6** NAME _____

Mrs. Bonny Tighe 25 points

11.1 – 11.3 TA _____ Wed 3/27/2024

1. Find the general term for the sequence Find the limit of the sequence and state if the sequence converges or diverges. Is the sequence bounded? If so, what are the bounds. Determine whether the sequence is increasing, decreasing or not monotonic

2. Determine whether the sequence converges or diverges. If it converges, find the limit.

a) $a_n = \ln(n+2) - \ln(3n+1)$ b) $a_n = \dfrac{4n-3}{3n+5}$

3. Fill in the blank: a) If $\sum_{k=1}^{\infty} a_k \ converges, \ then \ \lim_{k\to\infty} a_k =$ _____

b) What is the symbol for the 30th partial sum of a series?_____

c) If $\sum_{k=1}^{\infty} a_k \ diverges \ and \ \sum_{k=1}^{\infty} b_k \ converges \ then \ \sum_{k=1}^{\infty} (ca_k + db_k)$ _____

d) If $\lim_{k\to\infty} a_k = 0 \ then \ \sum_{k=1}^{\infty} a_k$ converges. True or False _____

4. a. State the test for Divergence

b. State the Geometric Series Test

c. State the Integral Test

d. State the p-series test

5. Determine whether the series is convergent or divergent and if it is convergent **find the sum**.

a) $\sum_{k=1}^{\infty} \frac{1}{k^2 + 2k}$

b) $\sum_{n=1}^{\infty} \frac{2^{n+2}}{3^n}$

6. Find the series. $2 + \frac{2 \cdot 5}{3!} + \frac{2 \cdot 5 \cdot 8}{6!} + \frac{2 \cdot 5 \cdot 8 \cdot 11}{9!} + \ldots$

7. Express $0.0\overline{35}$ as a ratio of integers using infinite series.

8. Use the **Integral test** to determine if the series is divergent or convergent.

$$\sum_{n=1}^{\infty} \frac{e^{1/n}}{n^2}$$

9. a) Estimate the error in using the partial sum S_{20} of the series $\sum_{n=1}^{\infty} \frac{1}{n^3}$

b) What partial sum would be necessary for the above series to have 0.001 accuracy?

MATH 152 **QUIZ 6A** NAME _____

Mrs. Bonny Tighe 25 points

11.1 – 11.3 TA _____ Wed 3/27/2024

1. Find the general term for the sequence $\frac{3}{1}, \frac{6}{2}, \frac{9}{6}, \frac{12}{24}$ Find the limit of the sequence and state if the sequence converges or diverges. Is the sequence bounded? If so, what are the bounds. Determine whether the sequence is increasing, decreasing or not monotonic

2. Determine whether the sequence converges or diverges. If it converges, find the limit.

a) $a_n = \dfrac{e^n}{n^3 + 2n + 3}$

b) $a_n = \ln(2n - 1) - \ln(n + 1)$

3. Fill in the blank: a) If $\sum_{k=1}^{\infty} a_k \ converges, \ then \ \lim_{k \to \infty} a_k =$ _____

b) What is the symbol for the 20th partial sum of a series?_____

c) If

$\sum_{k=1}^{\infty} a_k \ converges \ and \ \sum_{k=1}^{\infty} b_k \ converges \ then \ \sum_{k=1}^{\infty} (ca_k + db_k)$ _____

d) If $\lim_{k \to \infty} a_k = 0 \ then \ \sum_{k=1}^{\infty} a_k$ converges. True or False _____

4. a. State the Geometric Series Test

b. State the test for Divergence

c. State the Integral Test

d. State the p-series test

5. Determine whether the series is convergent or divergent and if it is convergent **find the sum**.

a) $\sum_{k=1}^{\infty} \frac{1}{k^2 + 4k + 3}$

b) $\sum_{n=0}^{\infty} \frac{3^{n+1}}{5^{n+2}}$

6. Find the series. $2 + \frac{2 \cdot 4}{2!} + \frac{2 \cdot 4 \cdot 6}{4!} + \frac{2 \cdot 4 \cdot 6 \cdot 8}{6!} + \ldots$

7. Express $0.\overline{35}$ as a ratio of integers using infinite series.

8. Use the **Integral test** to determine if the series is divergent or convergent.

$$\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$$

9. a) Estimate the error in using the partial sum S_{10} of the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$

b) What partial sum would be necessary for the above series to have 0.001 accuracy?